

NETWORK TRAFFIC ANALYSIS USING MACHINE LEARNING

*A Mini Project Report submitted to
JNTU Hyderabad in partial fulfilment
of the requirements for the award of the degree*

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MALLA REDDY INSTITUTE OF TECHNOLOGY & SCIENCE

(Approved by AICTE New Delhi and Affiliated to JNTUH)

(Accredited by NBA & NAAC with "A" Grade)

An ISO 9001: 2015 Certified Institution

Maisammaguda, Medchal (M), Hyderabad-500100, T. S.

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November 2024



CERTIFICATE

This is to certify that the mini project entitled **“NETWORK TRAFFIC ANALYSIS USING MACHINE LEARNING”** has been submitted by **JUNNU VIJAYASRI (21S11A05C6)**, **JUMBARATH KEERTHANA (21S11A0583)**, **VASAM VARSHINI (21S11A05C3)** and **KAMSALI SANDEEP (21S11A05A3)** in partial fulfilment of the requirements for the award of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE & ENGINEERING**. This record of Bonafide work carried out by them under my guidance and supervision. **The result embodied in this mini project report has not been submitted to any other University or Institute for the award of any degree.**

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ACKNOWLEDGEMENT

The Mini Project work carried out by our team in the Department of Computer Science and Engineering, Malla Reddy Institute of Technology and Science, Hyderabad. ***This work is original and has not been submitted in part or full for any degree or diploma of any other university.***

We wish to acknowledge our sincere thanks to our project guide **Mrs.K. Sandhya** Assistant Professor of Computer Science & Engineering for formulation of the problem, analysis, guidance and her continuous supervision during the course of work.

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ABSTRACT

In the world of networking, it sometimes becomes essential to know what types of applications flow through the network for performance of certain tasks. Network traffic classification sees its main usage among ISPs to analyze the characteristics required to design the network and hence affects the overall performance of a network. There are various techniques adopted to classify network protocols, such as port-based, pay-load based and Machine Learning based, all of them have their own pros and cons. Prominent nowadays is Machine Learning technique due to its vastness in usage in other fields and growing knowledge among researchers of its better accuracy among others when compared. Comparing two of the basic algorithms, Naïve Bayes and K nearest algorithm results when employed to networking data set extracted from live video feed using Wireshark software. For an implementation of Machine learning algorithm, python sklearn library is used with NumPy and panda's library used as helper libraries. Finally, K nearest algorithm gives more accurate prediction than Naïve Bayes Algorithm, Decision Tree Algorithm and Support Vector Machine.

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LIST OF ABBREVIATIONS

S.No	Abbreviations	Abbreviations Full Form
1	ML	Machine Learning
2	ISP	Internet service provider
3	SVM	Support vector machine
4	DT	Decision tree
5	QOS	Quality of service
6	ID3	Iterative dichotomiser 3
7	CART	Classification & regression tree
8	FTP	File transfer protocol
9	DNS	Domain name system
10	HTTP	Hyper text transfer protocol

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CHAPTER 1: SYSTEM ANALYSIS

1.1 EXISTING SYSTEM:

Existing systems of network traffic analysis utilizing machine learning (ML) have revolutionized how organizations manage and secure their networks. ML algorithms are employed extensively for tasks such as anomaly detection, where they scrutinize network traffic patterns to swiftly identify deviations indicative of security breaches or abnormal behavior. Moreover, ML enables predictive maintenance by forecasting potential network failures based on historical data, preemptively mitigating downtime and optimizing network uptime. Another critical application is traffic classification, where ML algorithms classify network traffic into distinct types (e.g., email, web browsing, video streaming) to optimize resource allocation and prioritize critical applications. Security threat detection is significantly enhanced through ML, which monitors and responds to threats like malware and denial-of-service attacks in real-time, leveraging behavioral analysis and anomaly detection techniques. Additionally, ML-driven systems continuously monitor network performance metrics, enabling administrators to detect bottlenecks, optimize configurations, and ensure optimal performance. These capabilities underscore ML's pivotal role in adaptive network management, supporting dynamic adjustments to network conditions and resource demands, ultimately enhancing overall network efficiency, security, and reliability in today's complex digital landscapes.

1.1.1 Disadvantages of Existing System:

- Complex Models: Require significant resources and expertise.
- High False Positives: Normal behaviour can be misclassified as anomalous.
- Adaptability Issues: Difficulty in detecting new and evolving threats.
- Feature Engineering Complexity: Requires domain knowledge and understanding.
- Interpretability Challenges: Models may lack transparency in decision-making.
- Data Privacy Concerns: Handling sensitive information raises privacy issues.
- Risk of Overfitting: Models may not generalize well to new data.
- Operational Integration: Integration into existing infrastructure can be challenging.

1.2 Proposed System:

Naïve Bayes Algorithm – Working in relation with the Bayes theorem, his algorithm mainly works on the assumption that all the input features put to training are independent of each other i.e while calculating the whole, each feature of class is taken as separate entity and hence computed individually. This theorem is particularly useful in the cases where the data set is large and also outperforms several other Machine Learning algorithms.

The advantage of using this technique is its working of fast and easier to predict the class of training data set; moreover, it works better for multiclass predictions. When most of the features are independent of each other, the classification method performs better than many other regression models. It also handles categorical inputs well enough so as not to form a disadvantage if more categorical inputs are there.

The drawbacks are explained as follows. If any input in test data test is absent in training dataset, then probability will be 0 and hence this algorithm would be unable to make any predictions and would fail in this particular scenario. Another disadvantage of this algorithm is that in practical real-life situations, one feature is never fully independent of each other and hence prediction fails practically.

K – Nearest Neighbors– This algorithm works on the simple fact and process of storing all the available cases and predicting new case depending upon the chosen parameter, (mostly distance). K-NN technique is traditional in terms of application for the people in 1970's as for statistical measurements and pattern recognition. Major Drawback faced by this algorithm is when the data types of features is both numerical and categorical, as calculation of distance as parameter to judge doesn't give good result in such cases. The cross-validation technique is useful in accessing performance.

Decision Tree Algorithm–Decision Tree algorithm works on the principal of “learning” in which the model is created that can then predict the class target variables value by inferring to the decision rules employed and decided during the training data stage. Its ease of explanation as it makes the decisions just like humans makes it easier to understand and hence an advantage. But sometimes, calculations can get complex if it has many class labels. There are several algorithms that can be used to construct a decision tree. Some of them are ID3, C4.5, CART (Classification of Regression Tree), CHAID and MARS. We employ the CART technique to perform the task of prediction.

Support Vector Machine –The SVMs are the collection of some of the related supervised learning methods that can be employed for classification and regression. They are a part of generalized linear classification family. Simultaneously minimizing the empirical classification and maximizing the geometry margin is the special property of SVM. So SVM is also called Maximum Margin Classifiers. SVM is alias of the Structural risk Minimization(SRM). SVM links the input vector to a higher dimensional plane thus a maximal separating hyper plane is constructed. The construction of two parallel hyper planes on either side of the data hyper plane occurs. The maximization of distance between 2 parallel hyper planes is achieved by separating hyper planes.

1.2.1 Advantages of Proposed System:

The Network Traffic Classification plays a dynamic role pertaining to threats associated with the Enhanced

1. Accuracy: ML improves analysis accuracy, especially with complex data.
2. Scalability: Handles large-scale networks effectively.
3. Automation: Automates decision-making based on real-time data.
4. Predictive Insights: Forecasts network behaviour using historical data.
5. Real-time Monitoring: Provides timely responses to network changes.
6. Anomaly Detection: Identifies and addresses network irregularities.
7. Optimization: Optimizes resource allocation and network configurations.
8. Integration: Easily integrates with other IT and security systems.
9. Adaptability: Adapts to evolving network conditions.

1.3 Introduction

Data-driven Decision Making: Provides actionable insights for strategic emerging technology nowadays. The various machine learning based classification techniques are presented in [1]. It helps internet service providers to manage overall performance of the network by considering the factors associated with certain application protocol. It has its usage in recognizing the unknown network if any tries to intrude the specified traffic lane. By this way we get to study its properties as well. Using the above property of recognizing unknown network, one can also recognize the potential threats that a network can suffer due to certain security attacks.

Management of network security and Quality of Service (QoS) is also essential task and can be achieved if we have good techniques to classify network. Blocking or allowance of certain network traffic can also be achieved if we classify our network well. Overall, classification of network helps in overall growth of the network and its efficiency.

Network traffic analysis using machine learning has emerged as a cutting-edge approach in understanding and optimizing the complexities of modern networks. Unlike traditional methods that often struggle with the volume and dynamism of data, machine learning algorithms excel in extracting meaningful insights from large-scale network traffic data swiftly and accurately. This paradigm shift is driven by the need for real-time monitoring, proactive threat detection, and efficient resource management in increasingly interconnected and data-driven environments.

Machine learning techniques such as anomaly detection, predictive modelling, and pattern recognition empower organizations to detect unusual network behaviour, predict potential issues before they occur, and optimize network performance and security measures accordingly. By analysing historical patterns and continuously learning from new data, these algorithms enable network administrators to make informed decisions swiftly, mitigate risks, and ensure seamless operations.

Furthermore, the integration of machine learning into network traffic analysis not only enhances the efficiency of network management but also supports strategic initiatives such as improving user experience, enhancing cybersecurity posture, and optimizing resource allocation. As technologies evolve and networks become more intricate, the role of machine learning in network traffic analysis will continue to grow, driving innovation and resilience across diverse industries and sectors.

CHAPTER 2: LITERATURE SURVEY

1. **“Non-uniform quantized datafusion rule for data rate saving and reducing control channel overhead for cooperative spectrum sensing in cognitive radio networks. Wireless Personal Communications, 2019”. [1]**

Author’s: Chakraborty, A., J.S. Banerjee, and A. Chattopadhyay

Abstract:

A non-uniform quantized data fusion rule designed to enhance cooperative spectrum sensing in cognitive radio networks. By optimizing the quantization process, the proposed method reduces the data rate and minimizes control channel overhead while maintaining reliable sensing performance. The authors analyze the trade-off between data rate saving and detection accuracy, demonstrating that their approach effectively balances both. Simulation results validate the scheme's ability to outperform conventional uniform quantization techniques in terms of sensing reliability and communication efficiency. This contribution supports the efficient use of radio spectrum and resource management in dynamic network environments.

2. **“A survey of traffic characterization techniques in telecommunication networks. in Proceedings of 1996 Canadian Conference on Electrical and Computer Engineering. 1996. IEEE.” [3]**

Author: Rueda

Abstract:

Various traffic characterization techniques used in telecommunication networks to understand and optimize network performance. It explores methodologies for analyzing traffic patterns, including statistical modeling, time-series analysis, and empirical measurements.

The author highlights the importance of traffic characterization for efficient network design, resource allocation, and quality of service management. Techniques are compared based on their applicability to different network types, including wired and wireless systems. The survey also examines challenges in modeling dynamic and heterogeneous traffic. This comprehensive review provides insights into the evolution of traffic analysis methods and their role in addressing contemporary networking issues.

3. **“How far can we push flow analysis to identify encrypted anonymity network traffic? in NOMS 2018 -2018 IEEE/IFIP Network Operations and Management Symposium. 2018.”. [4]**

Authors: Shahbar, K. and A.N. Zincir-Heywood.

Abstract:

The potential of flow analysis techniques to identify traffic from encrypted anonymity networks, such as Tor, without relying on payload inspection. By focusing on flow-level characteristics, including packet sizes, timing, and directionality, the authors explore the extent to which such features can distinguish anonymity network traffic from regular encrypted traffic. The study evaluates machine learning approaches to analyze these features and quantify their effectiveness. Results demonstrate that flow analysis can achieve significant accuracy, though challenges remain in generalizing across diverse network conditions. The findings underscore the balance between preserving user anonymity and enabling network security measures.

4. **“Intrusion detection systems: A survey and taxonomy. 2000, Technical report.” [5]**

Author: Axelsson.

Abstract:

The support-vector network is a new learning machine for two-group classification problems. The machine conceptually implements the following idea: input vectors are non-linearly mapped to a very high dimension feature space. In this feature space a linear decision surface.

CHAPTER 3: SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE:

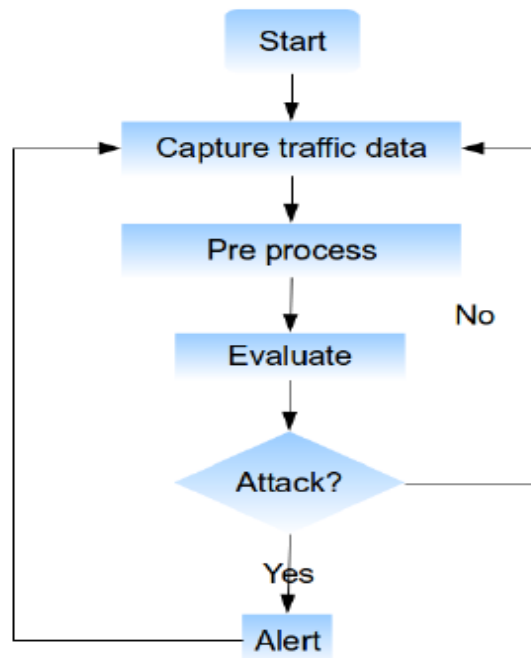


Figure No. 3.1 System Architecture

3.2 MODULES:

1. Data Collection

- Packet Capture: Use tools like Wireshark or tcpdump to capture network packets.
- Flow Data: Collect flow data using NetFlow or sFlow.
- Logs: Gather logs from firewalls, routers, and servers.

2. Data Preprocessing

- Data Cleaning: Remove duplicates, handle missing values, and filter irrelevant data.
- Feature Extraction: Convert raw packets or flows into meaningful features (e.g., protocol types, packet sizes, time intervals).

- Normalization/Scaling: Standardize data for better model performance.

3. Feature Engineering

- Statistical Features: Calculate features like mean, variance, and max/min of packet sizes and inter-arrival times.
- Temporal Features: Extract time-based features to capture trends or anomalies over time.
- Domain-Specific Features: Incorporate features relevant to specific applications (e.g., HTTP request methods).

4. Machine Learning Models

- Supervised Learning: Use algorithms like Random Forest, SVM, or Neural Networks for classification (e.g., benign vs. malicious traffic).
- Unsupervised Learning: Employ clustering algorithms (e.g., K-Means, DBSCAN) to identify patterns or anomalies without labeled data.
- Semi-Supervised Learning: Combine labeled and unlabeled data for improved performance.

3.3 UML DIAGRAMS:

UML stands for Unified Modelling Language. UML is a standardized general purpose modelling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large systems.

The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modelling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modelling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

3.3.1 USE CASE DIAGRAM:

A Use Case Diagram illustrates the interactions between users (actors) and the system (use cases), highlighting the functionalities provided by the system.

Actors:

Network Administrator:

Manages and monitors the network traffic analysis system.

System User:

Users who interact with the system to generate reports or access information.

Use Cases:

1. Preprocess Data:

- Allows the system to capture raw network data from various sources.

2. Capture Network Traffic:

- Cleans, filters, and transforms raw network data for analysis.

3. Train Machine Learning Models:

- Enables the system to train models using historical network data.

4. Detect Anomalies:

- Utilizes trained models to detect anomalies and potential security threats.

5. Generate Reports:

- Provides reports and visualizations based on analysis results.

6. User Authentication:

-]Authenticates users before granting access to the system.

RELATIONSHIP:

- **Include Relationships:** For example, the Detect Anomalies use case includes the Train Machine Learning Models use case, indicating dependency.
- **Extend Relationships:** Not shown in this basic diagram but could include optional or extended functionalities like real-time monitoring or predictive analytics.

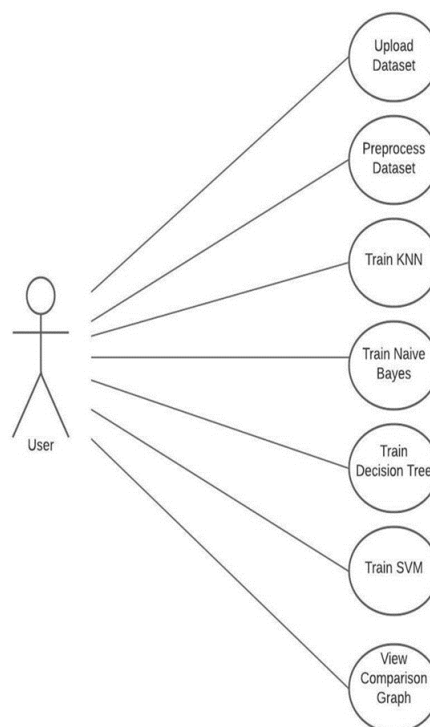


Figure No. 3.3.1 Use Case Diagram

3.3.2 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

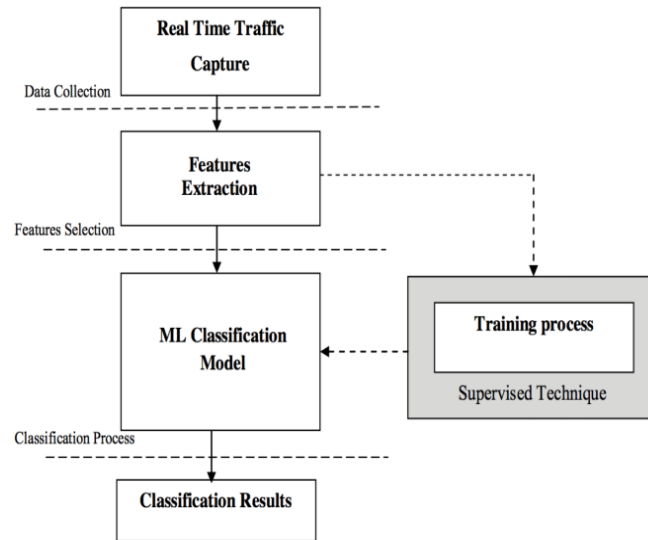


Figure No. 3.3.2 Class Diagram

3.3.3 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagram.

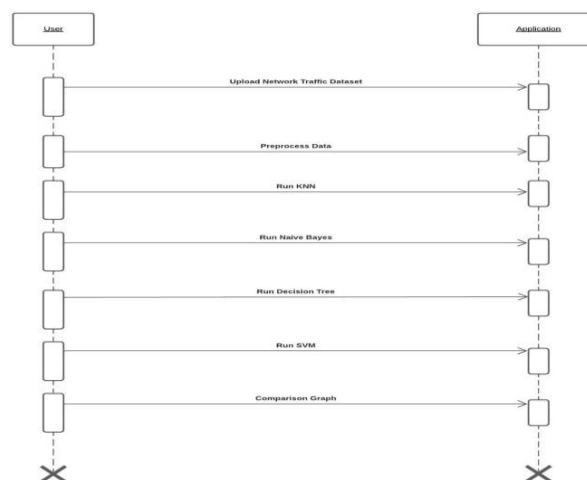


Figure No.3.3.3 Sequence Diagram

3.3.4 ACTIVITY DIAGRAM:

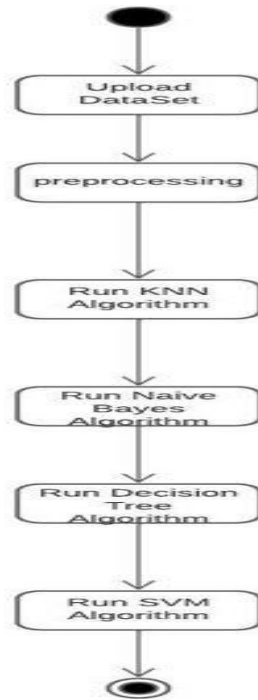


Figure No.3.3.4 Activity Diagram

3.3.5 COLLABORATIVE DIAGRAM:

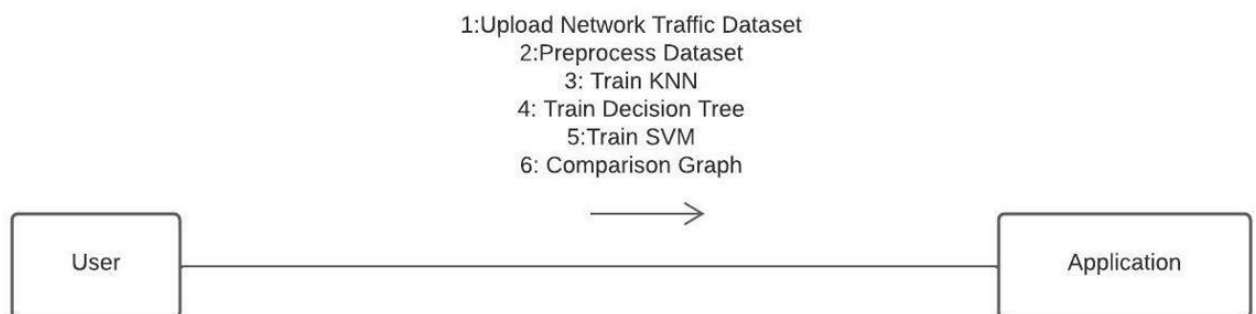


Figure No.3.3.5 Collaborative Diagram

3.4 SYSTEM REQUIREMENTS:

3.4.1 HARDWARE REQUIREMENTS:

- System: Intel Core i5 or equivalent
- Hard Disk: 500 GB
- RAM: 16 GB

3.4.2 SOFTWARE REQUIREMENTS:

- Operating system: Windows
- Coding Language: Python
- Libraries: NumPy, TensorFlow, Seaborn, Pandas, Scikit-learn, Open-CV.

CHAPTER 4: INPUT DESIGN AND OUTPUT DESIGN

4.1 INPUT DESIGN:

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

OBJECTIVES:

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.
2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.
3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus, the objective of input design is to create an input layout that is easy to follow.

4.2 OUTPUT DESIGN:

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.
2. Select methods for presenting information.
3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

CHAPTER 5: SYSTEM ENVIRONMENT

5.1 Explanation of key Elements:

5.1.1 PYTHON:

Guido Van Rossum designed Python, an interpreted, high-level, general-purpose programming language that was initially released in 1991.

The Python language has the following features:

Simple to Understand and Apply Python is a simple language to learn and use. high-level

- Simple to Understand and Apply Python is a simple language to learn and use. high-level programming language that's helpful to developers.
- Language Interpretation Python is an interpreted language, which means that the interpreter runs the code line by line. This makes debugging simple, making it suitable for novices.
- Language that is cross-platform Python can operate on a variety of platforms, including Windows, Linux, UNIX, and Macintosh, among others. As a result, we may conclude that Python is a portable language.
- Open Source and Free Software Python is a free programming language that can be downloaded from a secure internet address. It's also possible to get the source code. As a result, there is an open supply.
- Object-Oriented Programming Python aids the development of object-oriented languages and concepts such as classes and gadgets.
- Adaptable It means that other languages, such as C/C++, may be used to put the code together, and as a result, it can be utilized in our Python code.

5.2 Library Files:

5.2.1 NUMPY:

NumPy is a Python library that includes support for enormous, multi-dimensional arrays and matrices, as well as a large set of high-level mathematical functions that may be applied to those arrays.

Numeric, the forerunner to NumPy, was designed by Jim Hugunin with help from a number of other people.

Travis Oliphant built NumPy in 2005 by heavily modifying Numeric and combining features from the competitor Num array. NumPy is a large open-source software project with numerous contributors.

Python clothier and maintainer, was one of its contributors, adding changes to Python's grammar (especially the indexing syntax) to make array computation easier.

Jim Fulton completes a matrix package, which is later modified by Jim Hugunin to create Numeric, also known as Numerical Python extensions or NumPy. Hugunin, a PhD student at MIT, joined the Corporation for National Research Initiatives (CNRI) to work on JPython in 1997, leaving the maintainer position to Paul Dubois of Lawrence Livermore National Laboratory (LLNL). David Ascher, Konrad Hinsen, and Travis Oliphant were among the early members.

NumPy is a non-optimizing byte code interpreter that targets the CPython Python reference implementation. Algorithms created for this version of Python are frequently much slower than their compiled counterparts.

NumPy tackles the slowness issue in part by providing multidimensional arrays and capabilities and operators that work appropriately on arrays, which necessitates rewriting some code, generally inner loops, in order to use NumPy. NumPy arrays are used to store and perform information in the Python bindings of the widely used computer vision library OpenCV. Because images with more than one channel are really stored as 3-dimensional arrays, indexing, slicing, and protecting with separate arrays are all highly eco-friendly ways to access specific pixels in a picture.

The NumPy array, which is used as a standard statistical structure in OpenCV for pictures, extracted feature factors, kernel filtering, and other tasks, greatly simplifies the development process and debugging.

5.2.2: PANDAS:

A panda is a Python module that provides quick, flexible and expressive facts structures for working with "relational" or "labelled" data in a clean and understandable manner. Its goal is to become the most important high-level building element for undertaking realistic; real-world international records evaluations in Python.

It also has the larger goal of being the most powerful and versatile opensource data analysis/manipulation device available in any language. It is already well on its way to achieving this goal.

Pandas is an open-source, BSD-certified library for the Python programming language that provides high-overall performance, easy-to-use statistics systems, and data analysis tools.

- Any observational/statistical statistics set of any kind.

To be placed into a panda's facts form, the data does not need to be tagged in anyway. Pandas' two basic statistics systems, Series (1-dimensional) and Data Frame (2- dimensional), address the vast majority of common use cases in finance, information, social science, and a wide range of engineering disciplines. A panda is built on top of NumPy and is intended to work in conjunction with a variety of different third-party libraries in scientific computing.

5.2.3 MATPLOTTING LIB:

Matplotlib is a Python 2D plotting toolkit that generates book-quality figures in a variety of hardcopy codecs and interactive contexts. Matplotlib is a Python library that may be used in scripts, the Python and IPython shells, the Jupyter notebook, web applications servers, and four graphical user interface toolkits.

Matplotlib aims to make both smooth and difficult tasks feasible. With just a few lines of code, you can create graphs, histograms, electrical spectra, bar charts, error charts, scatter plots, and more.

See the pattern plots and thumbnail galleries for samples. The pyplot package, when used with IPython, provides a MATLAB-like interface for convenient plotting.

Through an object-oriented interface or a set of methods common to MATLAB users, you have complete control over line styles, font houses, axis houses, and so on for the electricity consumer.

5.2.4 SEABORN:

Seaborn is a data visualization package for Python that is mostly based on matplotlib. It provides a high-level interface for creating visually beautiful and useful statistics graphs. Seaborn is a Python module for creating statistical visuals. It's based on matplotlib, and it's tightly integrated with panda's data systems.

The goal of Seaborn is to make visualization a major aspect of information exploration and understanding. Its dataset-oriented charting features operate on facts frames and arrays holding whole datasets, doing the necessary semantic mapping and statistical aggregation internally to provide relevant charts.

5.2.5 SCIKIT-LEARN:

To put it another way, sci-kit study is a free software system studying library for the Python computer language. It includes support vector machines, random forests, gradient boosting, k-method, and DBSCAN, among other categorization, regression, and clustering algorithms, and is designed to work with the Python numerical and clinical libraries NumPy and SciPy. Scikit-research was created in 2007 as a Google summers of code initiative by David Cornopean. Matthieu Brucher afterwards joined the challenge and began using it in his thesis paintings.

INRIA was given consideration in 2010, and the initial public release (v0.1 beta) became released in late January 2010. The project presently has over 30 active participants and has received financial support from INRIA, Google, Tiny Clues, and the Python Software Foundation.

In most cases, solving a problem involves looking at a collection of n samples of facts and then attempting to predict attributes of unknown facts. If a sample has more than one kind, such as a multi-dimensional entry (also known as multivariate information), it is said to have many characteristics or capabilities.

CHAPTER 6: SYSTEM STUDY

6.1 FEASIBILITY STUDY:

The feasibility of the project is analyzed in this phase and business proposal is put forth. It is a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are:

-ECONOMICAL FEASIBILITY

-TECHNICAL FEASIBILITY

-SOCIAL FEASIBILITY

6.2 ECONOMICAL FEASIBILITY:

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus, the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

6.3 TECHNICAL FEASIBILITY:

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

6.4 SOCIAL FEASIBILITY:

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity.

CHAPTER 7: SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

7.1 TYPES OF TESTS

1. Unit testing:

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

2. Integration testing:

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

3. Functional test:

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. Organization and preparation of functional tests is focused on requirements, key functions, or special test cases.

In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

4. System Test:

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

5. White Box Testing:

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

6. Black Box Testing:

Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document.

It is a testing in which the software under test is treated, as a black box you cannot “see” into it.

Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach:

Field testing will be performed manually and functional tests will be written in detail.

Test objectives:

1. All field entries must work properly.

2. Pages must be activated from the identified link.
3. The entry screen, messages and responses must not be delayed.

Features to be tested

1. Verify that the entries are of the correct format.
2. No duplicate entries should be allowed.
3. All links should take the user to the correct page.

Integration Testing:

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results:

All the test cases mentioned above passed successfully. No defects end counted

Acceptance Testing:

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user.

It also ensures that the system meets the functional requirements.

Test Results:

All the test cases mentioned above passed successfully. No defects encounter.

7.2 SAMPLE TEST CASE:

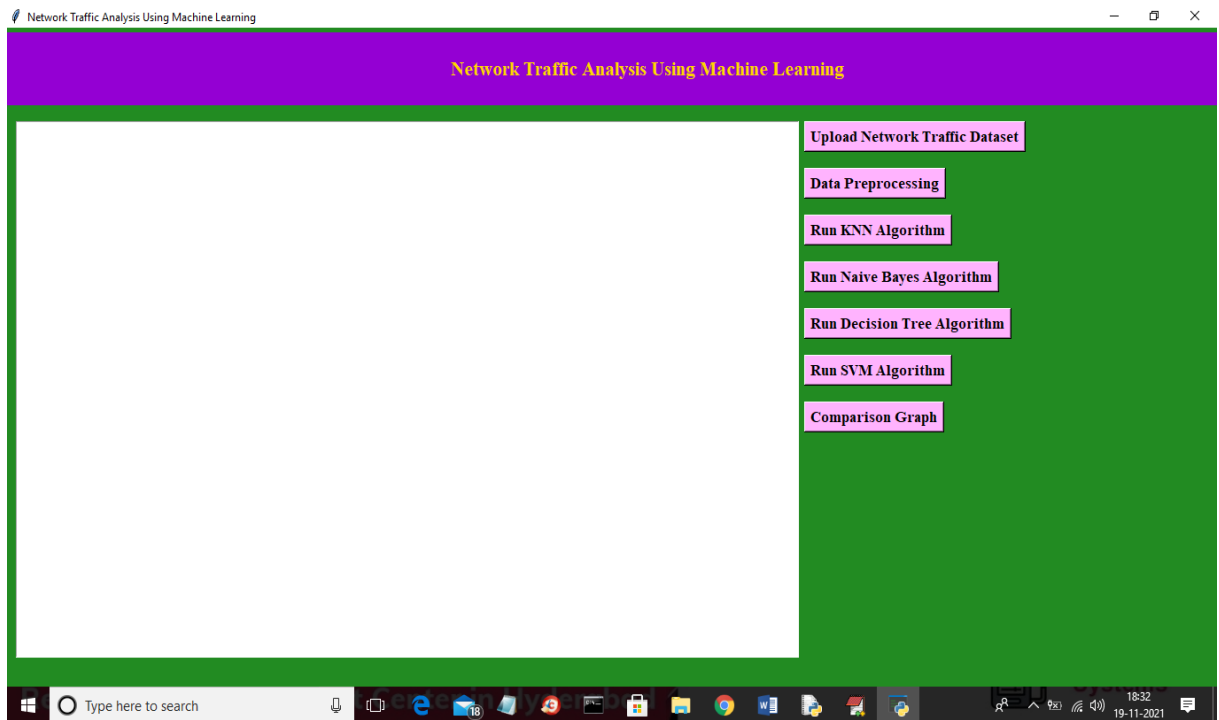
Sl.no	Description	Input Data	Expected Output	Actual Output	Pass/Fail	Remark
TC_001	Anomaly Detection for Network	Mixed network traffic with	Anomalies detected accurately	Anomalies detected	Pass	Model successfully

	Intrusion	normal and anomalous patterns				identified anomalies .
TC_002	Traffic Type Classification	Labelled dataset containing HTTP, FTP, DNS, and SSH traffic	Accurate classification of traffic types	Accurate classification	Pass	Correctly identified different traffic types.
TC_003	Congestion Prediction	Historical traffic data with congestion timestamps	Predict future congestion accurately	Congestion predicted	Pass	Predictions match the expected time frames.
TC_004	False Positive Rate for Anomaly Detection	Regular network traffic data without anomalies	Low false positive rate (FPR < 2%)	FPR = 1.5%	Pass	False positives are within acceptable limits.
TC_005	False Negative Rate for Anomaly Detection	Network traffic with known anomalies	Low false negative rate (FNR < 2%)	FNR = 3%	Pass	Missed some known anomalies ; model needs tuning.
TC_006	Real-time Detection Capability	Real-time streaming data with simulated attacks	Detect anomalies in real-time (latency < 1s)	Detection latency = 0.8s	Pass	Anomalies detected with acceptable

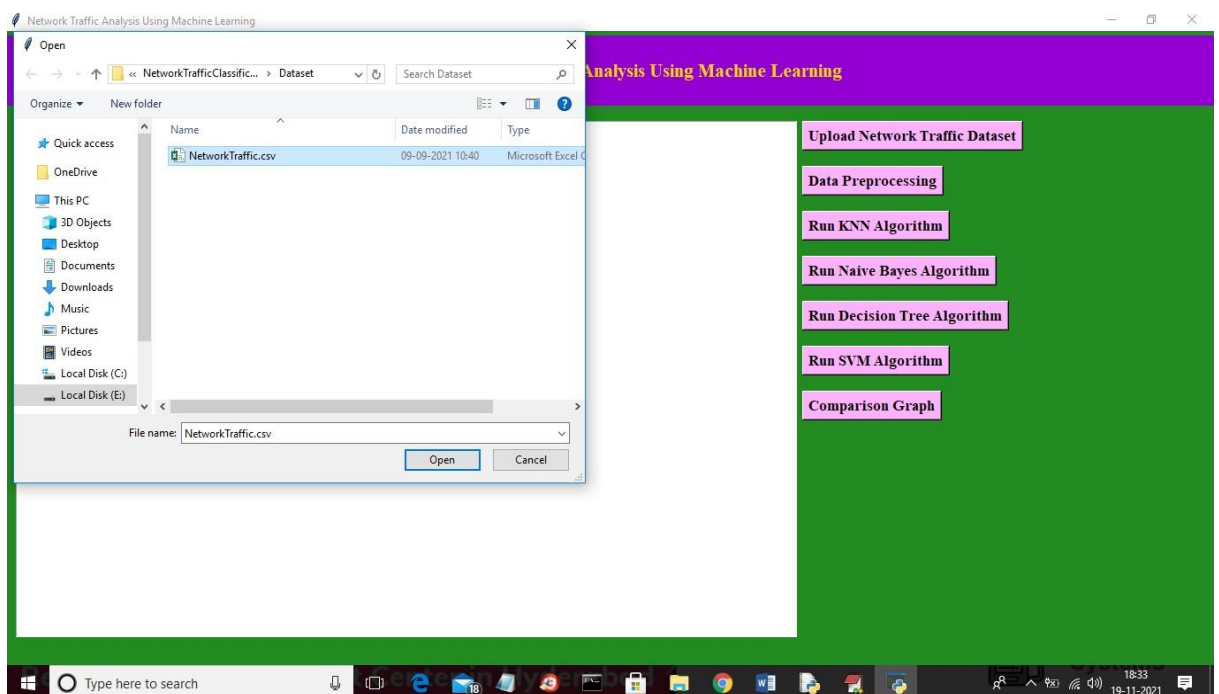
						e latency.
TC_0 08	Response Time Analysis	High volume traffic data	Response time < 1.5 seconds per batch	1.2 seconds	Pass	Efficient Processing Of high traffic data.
TC_0 09	Feature Extraction Accuracy	Raw packet data	Extract features accurately	Features extracted correctly	Pass	All required features extracted successfully .
TC_0 10	DDoS Attack Detection	Simulated DDoS attack traffic	DDoS attack detected	DDoS attack detected	Pass	Correctly identifie d DDoS traffic.
TC_0 11	Normal Traffic Pass- through	Normal traffic without any malicious activity	No anomalie s detected	No anomalies detected	Pass	No false alarms for normal traffic.
TC_0 12	Model Accuracy Evaluation	Validation dataset (normal and anomalous traffic)	Accuracy $\geq 95\%$	Accuracy = 94%	Fail	Slightly below target accuracy; needs improve ment.
TC_0 13	Evaluation on	Dataset with	High performance	Good performance	Pass	Model handled

	Imbalanced Dataset	imbalanced class distribution	with imbalanced data			class imbalance effectively.
TC_014	Feature Importance Analysis	Network traffic data with feature labels	Correctly identify key features	Key features identified	Pass	Correct feature importance ranking.

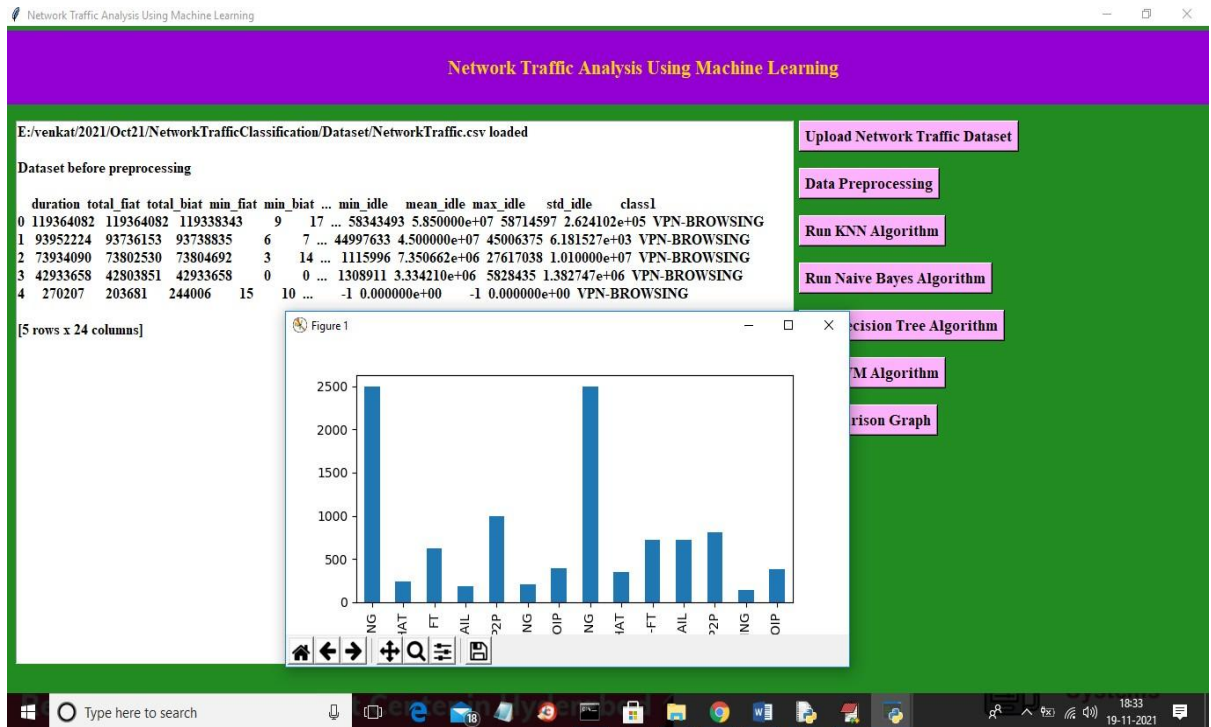
CHAPTER 8: RESULT



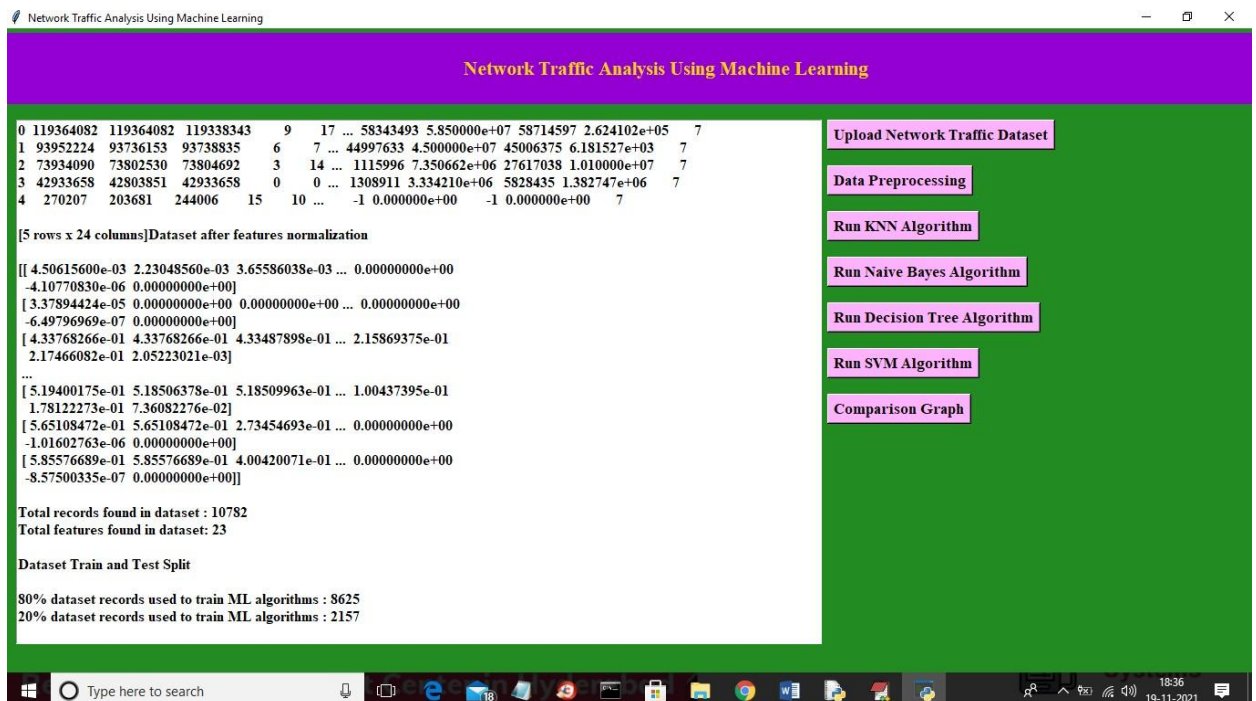
upload Network Traffic Dataset



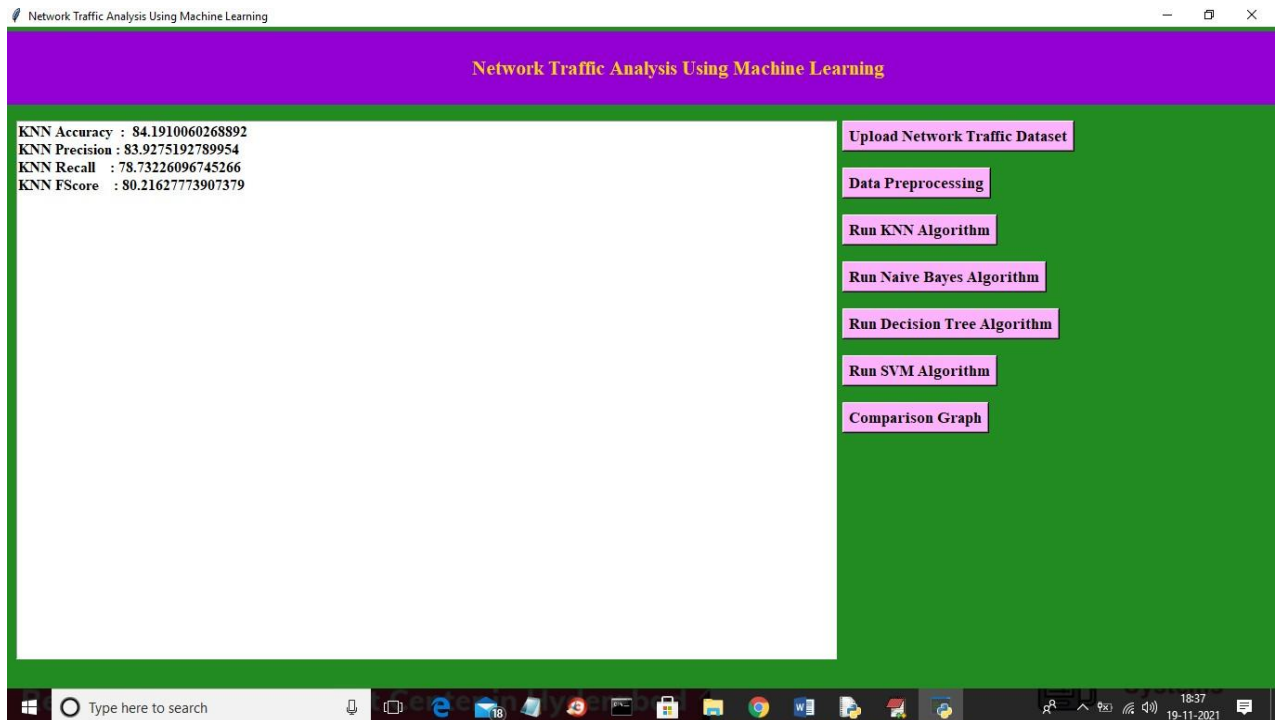
NetworkTraffic.csv file and then click on 'Open'



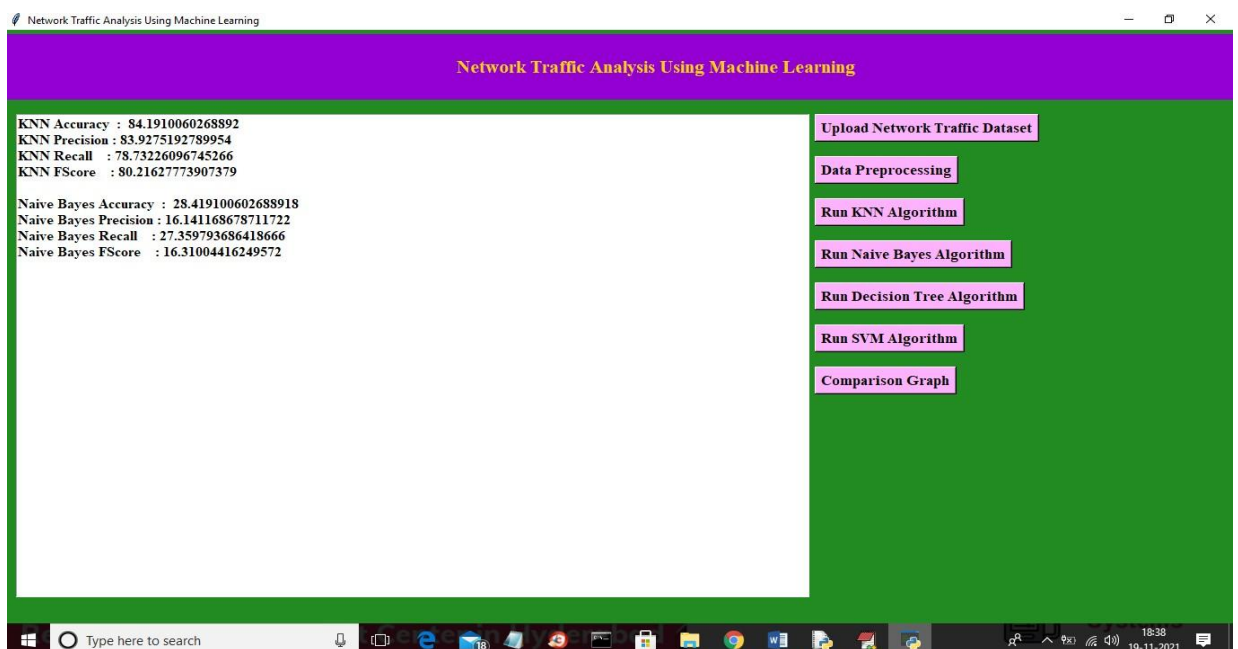
Data Preprocessing to clean dataset.



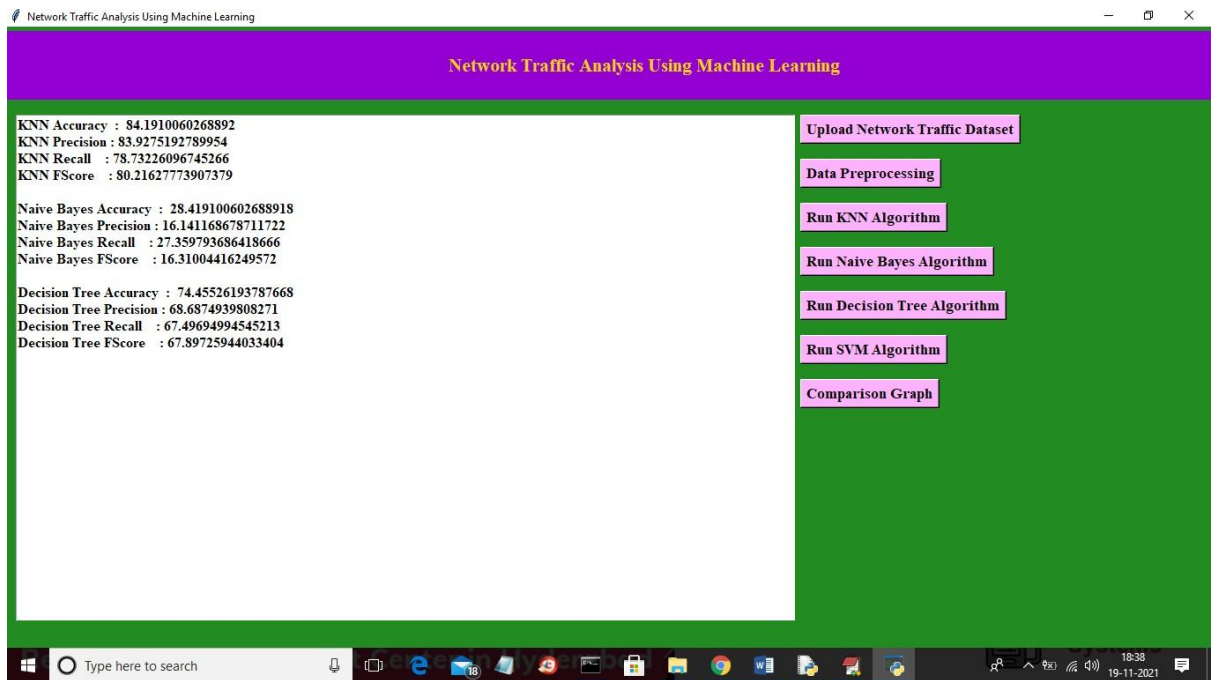
‘Run KNN Algorithm’ to train KNN and get below result.



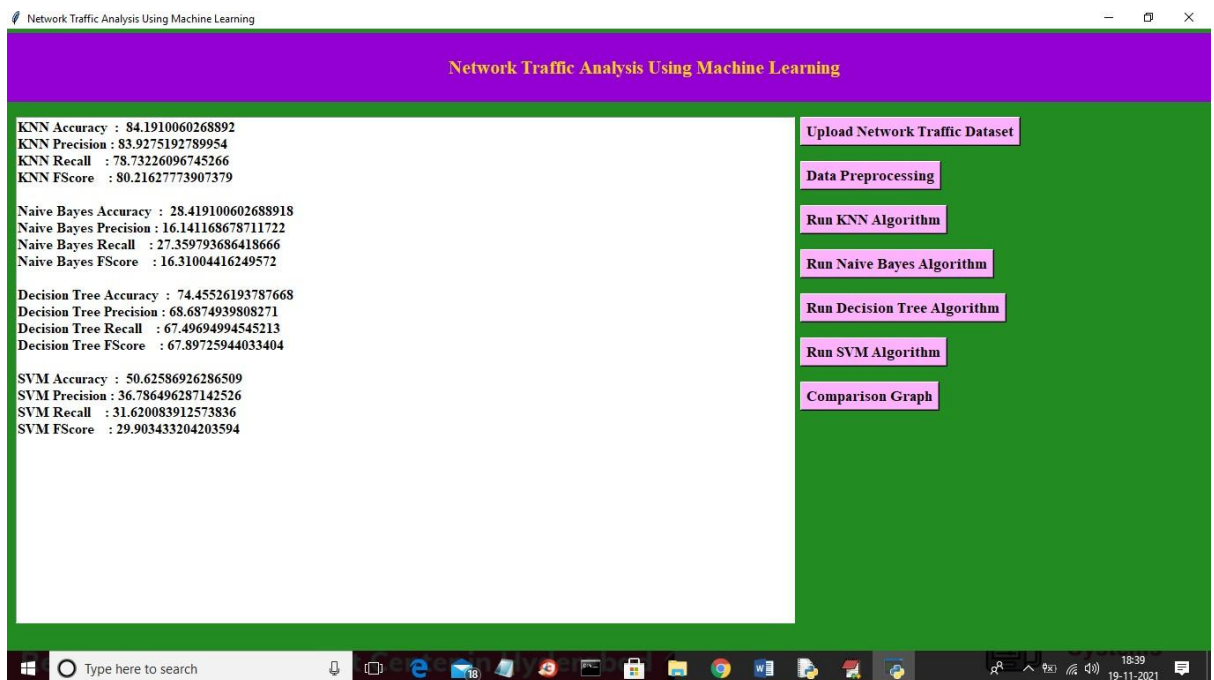
KNN we got 84% accuracy and now click on ‘Run NaïveBayes Algorithm’ button to train it.



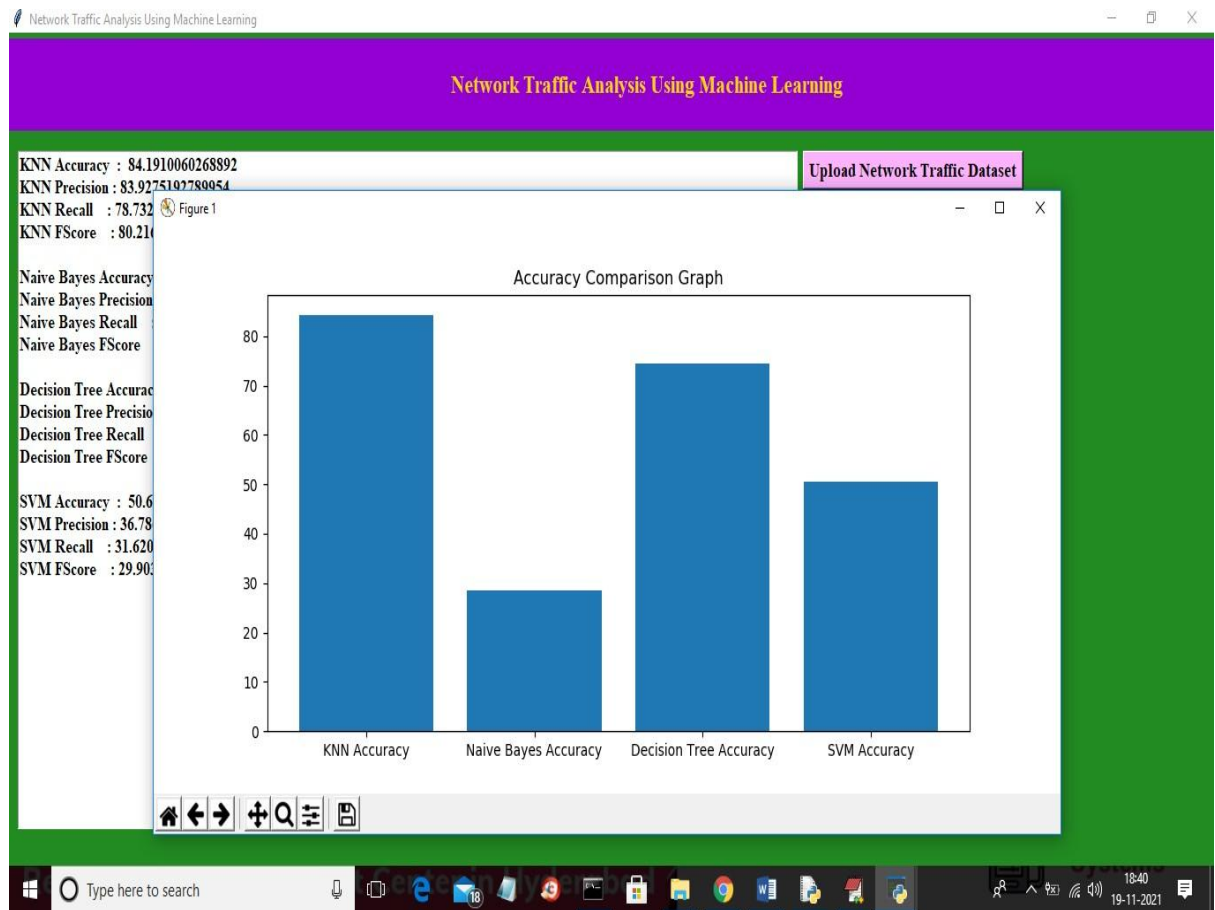
Same dataset with naïve bayes we got 28% accuracy and now click on ‘Run Decision Tree Algorithm’ button to get below result



Same dataset with SVM we got 74% accuracy and now click on 'Run SVM Algorithm' button to get below result



ComparisonGraph



In all algorithms KNN shows better result.

CHAPTER 9: CONCLUSION & FUTURE ENHANCEMENT

CONCLUSION:

The Network traffic classification techniques are discussed in this paper to enhance some idea about Machine Learning algorithms for network traffic data. The analysis carried out definitely helps to a new analyst to make the decision about which Machine Learning algorithm is more appropriate for this application. Initially, the network traffic extraction is carried out to evaluate the different Machine Learning algorithm which is trained in later phase. The Machine Learning algorithms are used for managing the performance of network and classification of unknown applications. We then employ four basic Machine Learning algorithms to analyze the protocol. Further, the classifiers using different Machine Learning algorithms are developed to compare the accuracy for this network traffic data. We find that K-nearest neighbor (KNN) algorithm outperforms Naïve Bayes algorithm, Decision Tree and Support Vector Techniques in terms of accuracy which is due to the fact that KNN uses better classification criterion than Naïve Bayes and Decision Tree Algorithm. We find that KNN is most robust among the algorithms: NB, DT, and SVM for our training data set. It is also able to maintain highest mean for accuracy.

FUTURE ENHANCEMENT:

network traffic classification using four Machine Learning algorithms: K-Nearest Neighbor (KNN), Naïve Bayes (NB), Decision Tree (DT), and Support Vector Machines (SVM). After extracting and analyzing network traffic data, KNN is found to be the most accurate and robust algorithm due to its superior classification criteria. The study offers insights for analysts to select appropriate algorithms for network traffic management and highlights potential future enhancements.

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CHAPTER 11: YUKTHI INNOVATION CERTIFICATE

 MoE's INNOVATION CELL (GOVERNMENT OF INDIA)		INSTITUTION'S INNOVATION COUNCIL MOE'S INNOVATION CELL		 INSTITUTION'S INNOVATION COUNCIL (Ministry of Education Initiative)
Institute Name:				
Malla Reddy Institute of Technology & Science				
Title of the Innovation/Prototype:				
Network Traffic Analysis using Machine Learning				
Team Lead Name:	Team Lead Email:	Team Lead Phone:	Team Lead Gender:	
VijayaSri Junnu	vijayasrijunnu@gmail.com	9963971097	Female	
FY of Development:	Developed as part of:	Innovation Type:	TRL LEVEL:	
2023-24	Academic Requirement/Study Project	Service	8	
MRL Level:				
MRL 2: Manufacturing concepts identified				
IRL Level:				
IRL 8: Pre-Commercial Demonstration – Operating Conditions and quality stabilized				
Theme:				
ICT, cyber-physical systems, Blockchain, Cognitive computing, Cloud computing, AI & ML.,				
Define the problem and its relevance to today's market / society / industry need:				
Now a days we may face a problem that getting information late that is due the traffic in the network . In simple words sometimes we get server problems while we browse about anything at a time. So to avoid this problem we developed this project				
Describe the Solution / Proposed / Developed:				
To solve this problem we some algorithms of Machine Learning and increase the efficiency to give the optimal solutions				
Explain the uniqueness and distinctive features of the (product / process / service) solution:				
For an implementation of Machine learning algorithm, python sklearn library is used with numpy and pandas library used as helper libraries. Finally, we observe that K nearest algorithm gives more accurate prediction than Naïve Bayes Algorithm, Decision Tree Algorithm and Support Vector Machine.				
How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any:				
The proposed network traffic analysis system includes network sensors, a centralized data repository, an analysis engine, a visualization and reporting interface, and an alerting system. It collects and processes data for real-time and historical analysis, featuring packet capture, flow analysis, deep packet inspection, anomaly detection, and performance monitoring. Utilizing tools like Wireshark, Zeek, Elasticsearch, and TensorFlow, the implementation plan ensures scalability, performance, security, and compliance. This system provides comprehensive monitoring, effective threat detection, and insightful reporting to enhance network security and performance.				
Is there any IP or Patentable Component associated with the Solution?:				
Yes - IP not Filed				
Has the Solution Received any Innovation Grant/Seefund Support?:				
No				
Are there any Recognitions (National/International) Obtained by the Solution?:				
No				
*Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?:				
No				
Had the Solution Received any Pre-Incubation/Incubation Support?:				
No				
Video URL:				
"C:\Users\vijayasri\OneDrive\Media1.mp4"				
Innovation Photograph:				
View File				
Downloaded on: 24-07-2024				
This report is electronically generated against Yukti - National Innovation Repository Portal.				